

Three dimensions: sharpness of $h_\Gamma^{3/2}$ and general smooth obstacles (td:open items 4 and 5)

subagent report

notes/v6/td_sharp, 2026-09-28

Summary and status

Statement	Status
Thm 1.10: $\ \nabla(\tilde{u} - u_h)\ \geq c\kappa_0\ W\ _{L^2(\Gamma)}h_\Gamma^{3/2} - Ch_\Gamma^2$ for CNS* (and GS), sphere, under the 3D boundary condition (M3 ³)	[PROVED]
(M3 ³) on Alfeld-refined meshes, every $k \geq 4$, one macro-tetrahedron per face, uniform in the shape (Thm 1.14)	[PROVED] , computer-assisted, exact rational arithmetic (rank 8 of an 8×26 matrix)
(M3 ³) on any shape-regular tetrahedral mesh, $k \geq 9$ (Prop. 1.15)	[PROVED] , closed form
Hence: sharpness of Scott's rate for CNS* on Alfeld meshes, all $k \geq 4$ (Cor. 1.11)	[PROVED] (modulo the (IS3) uniformity caveat of td:open item 3, which is needed only for the <i>upper</i> bound)
$k = 3$ on Alfeld meshes: a single macro-element cannot work (Prop. 1.16); vertex fans of macro-elements do in generic exact tests	[PROVED] negative / [NUMERICAL] (exact rational ranks on 6 fans); (M3 ³) for $k = 3$ open
Part B: Thm td:main and Cor. td:ns on a general smooth (non-convex, several components) obstacle in 3D, inscribed triangulated surfaces, $\Omega \not\subset \Omega_h$, global mean (Thm 2.10)	[PROVED MODULO (IS3)] , exactly as in the sphere case; all domain ingredients proved (uniform Bogovskiĭ, Korn, extension, sagitta, test field)
Lower bound on general surfaces, curvature-weighted: $\ \nabla(\tilde{u} - u_h)\ \geq ch_\Gamma^{3/2}\ \mathfrak{h}\ W\ _{L^2(\Gamma)} - Ch_\Gamma^2$ (Thm 2.11)	[PROVED] for Alfeld $k \geq 4$ and all meshes $k \geq 9$

Two findings worth recording. (i) On a face inscribed in the sphere the sagitta is *exactly* the circum-sphere function: $d = \frac{1}{2} \sum_{i < j} \ell_{ij}^2 \mu_i \mu_j + O(h_\Gamma^4)$ (Lemma 1.6); on a general surface the weights become $\mathfrak{h}(e_{ij}, e_{ij})$ (second fundamental form on the edge vectors). The whole 3D lower bound reduces to the three *edge-bubble moments* $\int_F \mu_i \mu_j \partial_n v$, and on Alfeld splits their joint surjectivity is an affine-invariant property of one reference macro-element (no compactness over shapes). (ii) For $k = 3$ the single-macro-element image is one-dimensional and equals $\sum_{\text{cyc}} \ell_{ij}^2 e_{ij}$, which vanishes exactly on equilateral faces (the 3D analogue of the centroid/circumcentre defect of Lemma td:faces(d)).

Throughout, the notation and conventions of paper Section 12 (td) are used: n is the outward normal of Ω_h on Γ_h (into P_h), $a_h(u, v) = \frac{1}{2}(Du, Dv)$, $N_h, \mathcal{G}, \|\cdot\|$ as in Sections 2, 5, 12. C, c depend on shape regularity, k , the geometry and norms of $(u, p, \tilde{u}, \tilde{p})$, never on $h, h_\Gamma, \mu, \gamma, \rho$.

1 Part A: the lower bound in 3D

1.1 Setting and the abstract lower bound

Setting of td:setting: $\Omega = R \setminus \overline{D}$, D the unit ball, P_h a convex inscribed polyhedron with triangular faces, $\mathcal{T}_h$ a conforming shape-regular mesh of $\Omega_h = R \setminus P_h$ whose boundary faces $F \in \mathcal{F}_h^\Gamma$ are whole faces of P_h , (M0³), (D). For $F \in \mathcal{F}_h^\Gamma$ let $K_F \in \mathcal{T}_h$ be the tetrahedron with face F .

Lemma 1.1 (One boundary face per tetrahedron). *For h_Γ small (in terms of shape regularity), no tetrahedron has two faces in Γ_h . In particular $F \mapsto K_F$ is injective.*

Proof. Two faces of K in Γ_h share an edge E of Γ_h , and the dihedral angle of K at E equals the dihedral angle of Ω_h at E , which is $\pi + \angle(n_F, n_{F'}) > \pi$ since P_h is convex. (On a general surface, Part B, it is $\pi + O(h_\Gamma)$, which exceeds the maximal dihedral angle $\pi - \delta(\text{shape reg.})$ for h_Γ small.) $\square$

Put $Y_h := Z_h \cap V_h^0$ (divergence-free, zero on $\Gamma_h \cup \partial R$) and

$$Y_h^1 := \left\{ v \in Y_h : \int_F \partial_n v \, dA = 0 \quad \forall F \in \mathcal{F}_h^\Gamma \right\}.$$

Lemma 1.2. *If $v \in Y_h$ and $F \in \mathcal{F}_h^\Gamma$, then on F (trace from K_F) $\nabla v = \partial_n v \otimes n_F$ and $\partial_n v \cdot n_F = 0$; $\partial_n v$ is tangential to F .*

Proof. $v|_{K_F}$ is a polynomial vanishing on the plane of F , so its tangential derivatives vanish there; then $0 = \text{div } v = \partial_n v \cdot n_F$. $\square$

Lemma 1.3 (Error identity; dimension-free). *Let u_h be the CNS* velocity ($\gamma \geq \gamma_2$, under (IS3)) or the GS velocity ($\gamma \geq \gamma_0$), $e_u = \tilde{u} - u_h$. For $v \in Y_h$: $a_h(e_u, v) - \langle e_u, \partial_n v \rangle = -\langle \tilde{u}, \partial_n v \rangle - (F, v)$.*

Proof. Verbatim Lemma lb:identity: the error equation (eq:cnserr) resp. Cor. cor:GSerr hold in 3D (Thm td:main); on Y_h the pressure form $B^*(e_p, v)$ and $\mathcal{R}(v)$ vanish ($\text{div } v = 0$, $v|_{\Gamma_h} = 0$), as do the transposed and penalty parts of $\mathcal{G}$. $\square$

Lemma 1.4 (Trace control on Y_h^1). *Let C_{PT} be the constant of $\|w - \bar{w}_K\|_{L^2(F)} \leq C_{PT}(\text{diam } K)^{1/2} \|\nabla w\|_K$ ($F \subset \partial K$, $\bar{w}_K$ the mean over K) and C_I that of $\|\nabla v\|_{L^2(F)} \leq C_I(\text{diam } K)^{-1/2} \|\nabla v\|_K$ for $v|_K \in P_k$. Then $|\langle w, \partial_n v \rangle| \leq C_{PT} C_I \|\nabla w\| \|\nabla v\|$ for $w \in H^1(\Omega_h)^3$, $v \in Y_h^1$.*

Proof. $\int_F w \cdot \partial_n v = \int_F (w - \bar{w}_{K_F}) \cdot \partial_n v$; Cauchy–Schwarz on each F and over the distinct K_F (Lemma 1.1). Both constants are scaling/shape-regularity constants in 3D. $\square$

Corollary 1.5 (Abstract lower bound). *With $\Lambda_0 := 2 + C_{PT} C_I$: $\Lambda_0 \|\nabla e_u\| \geq \sup_{0 \neq v \in Y_h^1} |\langle \tilde{u}, \partial_n v \rangle + (F, v)| / \|\nabla v\|$.*

Proof. $|a_h(e_u, v)| \leq 2 \|\nabla e_u\| \|\nabla v\|$ and Lemmas 1.3, 1.4. $\square$

1.2 The sagitta is the circumsphere function

For $F \in \mathcal{F}_h^\Gamma$ with vertices z_1, z_2, z_3 , barycentrics μ_1, μ_2, μ_3 , edge lengths $\ell_{ij} = |z_i - z_j|$, circumcentre o_F , circumradius r_F :

Lemma 1.6 (Sagitta). *For $x \in F$, $d(x) := \text{dist}(x, \Gamma) = 1 - |x|$ satisfies*

$$d(x) = \frac{1}{2} (r_F^2 - |x - o_F|^2) + \delta(x) = \frac{1}{2} \sum_{i < j} \ell_{ij}^2 \mu_i \mu_j + \delta(x), \quad 0 \leq \delta \leq r_F^4/4.$$

Proof. With $s := r_F^2 - |x - o_F|^2 \in [0, r_F^2]$, $|x|^2 = d_F^2 + |x - o_F|^2 = 1 - s$ (td:faces), and $1 - \sqrt{1 - s} \in [s/2, s/2 + s^2/4]$ for $s \leq \frac{1}{2}$. For $x = \sum \mu_i z_i$: $\sum_i \mu_i |z_i - o|^2 - |x - o|^2 = \sum_{i < j} \mu_i \mu_j |z_i - z_j|^2$ (variance identity), and $|z_i - o_F| = r_F$. $\square$

By Lemma lem:wall, $W := \partial_n u|_\Gamma$ is a tangential vector field on Γ (the wall shear); $\tilde{u}(x^*) = 0$ and $x = x^* + d(x)n(x^*)$ ($x^* = x/|x|$) give

$$\tilde{u}(x) = d(x) W(x^*) + O(d^2) \quad (x \in \Gamma_h), \quad |O(d^2)| \leq \|\tilde{u}\|_{W^{2,\infty}} d^2. \quad (1)$$

For $v \in Y_h$ and $F \in \mathcal{F}_h^\Gamma$ define the face moment (a vector parallel to F , Lemma 1.2)

$$\mathcal{M}_F(v) := \frac{1}{2} \sum_{i < j} \ell_{ij}^2 \int_F \mu_i \mu_j \partial_n v \, dA, \quad \ell_W(v) := \sum_{F \in \mathcal{F}_h^\Gamma} W_F \cdot \mathcal{M}_F(v), \quad W_F := W(x_F^*),$$

x_F the centroid.

Lemma 1.7 (Leading term). *For $v \in Y_h^1$: $|\langle \tilde{u}, \partial_n v \rangle - \ell_W(v)| \leq Ch_\Gamma^{5/2} \|\nabla v\|$ and $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$, C depending on $\|\tilde{u}\|_{W^{2,\infty}}$, $\|W\|_{W^{1,\infty}(\Gamma)}$, shape regularity.*

Proof. On F , by (1), Lemma 1.6 and $|W(x^*) - W_F| \leq Ch_\Gamma$: $\tilde{u} = (\frac{1}{2}(r_F^2 - |x - o_F|^2))W_F + R$ with $|R| \leq Ch_\Gamma^3$. Since $\int_F \partial_n v = 0$, the constant $\frac{1}{2}r_F^2$ drops, and $-\frac{1}{2}|x - o_F|^2 \equiv \frac{1}{2} \sum \ell_{ij}^2 \mu_i \mu_j$ modulo constants; so $\int_F \tilde{u} \cdot \partial_n v = W_F \cdot \mathcal{M}_F(v) + O(h_\Gamma^3) \|\partial_n v\|_{L^1(F)}$. Now $\|\partial_n v\|_{L^1(F)} \leq |F|^{1/2} C_I (\text{diam } K_F)^{-1/2} \|\nabla v\|_{K_F} \leq Ch_\Gamma^{1/2} \|\nabla v\|_{K_F}$ and $\#\mathcal{F}_h^\Gamma \leq Ch_\Gamma^{-2}$ ($|F| \geq ch_\Gamma^2$, $|\Gamma_h| \leq 4\pi$), so $\sum_F \|\partial_n v\|_{L^1(F)} \leq Ch_\Gamma^{-1/2} \|\nabla v\|$. The F -term is td:tools(b). $\square$

1.3 The local condition and the assembly

Assumption 1.8 ((M3³)). There are $\kappa_0 > 0$, K_{ov} , C_s such that for every $F \in \mathcal{F}_h^\Gamma$ and every unit vector t parallel to F there is $v_{F,t} \in Y_h^1$ with (i) $\|\nabla v_{F,t}\| = 1$; (ii) $\text{supp } v_{F,t} \subset \omega_F$, a union of tetrahedra within distance $C_s h_\Gamma$ of F , each tetrahedron lying in at most K_{ov} of the ω_F ; (iii) $t \cdot \sum_{F' \in \mathcal{F}_h^\Gamma} \mathcal{M}_{F'}(v_{F,t}) \geq \kappa_0 (\text{diam } F)^{5/2}$.

(iii) is scale invariant: under $x \mapsto \lambda x$ (with $v \mapsto v(\cdot/\lambda)$ renormalised) $\|\nabla v\|$ scales like $\lambda^{1/2}$ and $\mathcal{M}_F$ like λ^3 . The witnesses of Section 1.4 are supported in one tetrahedron or one macro-tetrahedron $T_F \supset F$, so $\omega_F \cap \Gamma_h = F$, $K_{\text{ov}} = 1$.

Lemma 1.9 (Assembly). *Assume (M3³) with $\omega_F \cap \Gamma_h = F$ (this is the case below; the general case only adds a term $Ch_\Gamma \|W\|_{W^{1,\infty}}$ to h_1 as in Lemma lb:assemble). Let $W \neq 0$ and $h_\Gamma \leq h_1$. For F with $W_F^\parallel$ (the projection of W_F on the plane of F) nonzero put $v_F := |W_F^\parallel| v_{F,t_F}$, $t_F = W_F^\parallel / |W_F^\parallel|$ (else $v_F := 0$), and $v := \sum_F v_F \in Y_h^1$. Then*

$$\ell_W(v) \geq \frac{1}{2} \kappa_0 c_0^{5/2} c_A^{1/2} K_{\text{ov}}^{-1/2} \|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} \|\nabla v\|,$$

where $c_A h_\Gamma^2 \leq |F|$ (shape regularity of the faces, c_A explicit from c_0 and the minimal angle).

Proof. 1. Since $\omega_F \cap \Gamma_h = F$, $\mathcal{M}_{F'}(v_F) = 0$ for $F' \neq F$ and $\ell_W(v) = \sum_F W_F \cdot \mathcal{M}_F(v_F)$. Write $W_F = W_F^\parallel + W_F^\perp$. Since $W_F \perp n(x_F^*)$ and $|n(x_F^*) - n_F| \leq Ch_\Gamma$ (td:faces(b)), $|W_F^\perp| \leq Ch_\Gamma |W_F|$; and $|\mathcal{M}_F(v_F)| \leq \frac{1}{2} r_F^2 |F|^{1/2} \|\partial_n v_F\|_{L^2(F)} \leq Ch_\Gamma^{5/2} \|\nabla v_F\|$. Since $\mathcal{M}_F(v_F)$ is parallel to F , $W_F \cdot \mathcal{M}_F(v_F) = W_F^\parallel \cdot \mathcal{M}_F(v_F)$, and by (M3³)(iii)

$$W_F \cdot \mathcal{M}_F(v_F) \geq \kappa_0 (\text{diam } F)^{5/2} |W_F^\parallel|^2 \geq \kappa_0 c_0^{5/2} h_\Gamma^{5/2} (1 - Ch_\Gamma^2) |W_F|^2.$$

2. *Riemann sum.* The radial projections F^* of the faces tile Γ with area Jacobian $1 + O(h_\Gamma^2)$ (td:faces(c)), and $||W|^2 - |W_F|^2| \leq Ch_\Gamma \|W\|_{W^{1,\infty}}^2$ on F^* ; so $\Sigma := \sum_F |F| |W_F|^2$ satisfies $|\Sigma - \|W\|_{L^2(\Gamma)}^2| \leq Ch_\Gamma \|W\|_{W^{1,\infty}}^2$. With $c_A h_\Gamma^2 \leq |F| \leq h_\Gamma^2$: $\Sigma/h_\Gamma^2 \leq \sum_F |W_F|^2 \leq \Sigma/(c_A h_\Gamma^2)$.
3. Hence $\ell_W(v) \geq \kappa_0 c_0^{5/2} (1 - Ch_\Gamma^2) h_\Gamma^{1/2} \Sigma$ and, by bounded overlap, $\|\nabla v\|^2 \leq K_{\text{ov}} \sum_F |W_F|^2 \leq K_{\text{ov}} \Sigma/(c_A h_\Gamma^2)$, so $\ell_W(v)/\|\nabla v\| \geq \kappa_0 c_0^{5/2} (1 - Ch_\Gamma^2) c_A^{1/2} K_{\text{ov}}^{-1/2} h_\Gamma^{3/2} \Sigma^{1/2}$. For $h_\Gamma \leq h_1$ (depending on $\|W\|_{W^{1,\infty}}/\|W\|_{L^2}$), $\Sigma \geq \frac{3}{4} \|W\|_{L^2}^2$ and $1 - Ch_\Gamma^2 \geq \frac{1}{\sqrt{3}}$; combine. $\square$

Theorem 1.10 (H^1 lower bound in 3D). *Assume $(M0^3)$, (D) , $(M3^3)$, and let $W = \partial_n u|_\Gamma \neq 0$. Let u_h be the velocity of CNS* ($\gamma \geq \gamma_2$; this needs (IS3) for existence) or of GS ($\gamma \geq \gamma_0$, any G). Then for $h_\Gamma \leq h_1$*

$$\|\nabla(\tilde{u} - u_h)\|_{\Omega_h} \geq \frac{\kappa_0 c_0^{5/2} c_A^{1/2}}{2\Lambda_0 K_{\text{ov}}^{1/2}} \|W\|_{L^2(\Gamma)} h_\Gamma^{3/2} - Ch_\Gamma^2,$$

with C independent of γ, μ, ρ, h , the pressure and $\varepsilon_h^{(3)}$.

Proof. Corollary 1.5 with v of Lemma 1.9; Lemma 1.7 gives $|\langle \tilde{u}, \partial_n v \rangle + (F, v)| \geq \ell_W(v) - Ch_\Gamma^2 \|\nabla v\|$. $\square$

Corollary 1.11 (Sharpness of Scott's rate in 3D). *On Alfeld-refined meshes with $k \geq 4$ (Thm 1.14), or on any mesh with $k \geq 9$ where CNS* is well posed (Prop. 1.15), if $W \neq 0$, $\gamma \in [\gamma_2, \gamma_{\max}]$ and $\varepsilon_h^{(3)} \leq Ch_\Gamma^{3/2}$: $ch_\Gamma^{3/2} \leq \|\nabla(\tilde{u} - u_h)\| \leq \|\tilde{u} - u_h\| \leq Ch_\Gamma^{3/2}$.*

The upper bound is Thm td:main(ii) and inherits its caveat (td:open item 3); the lower bound does not use (IS3) beyond existence of u_h .

1.4 Verification of $(M3^3)$

Transport of face data. Let T be a tetrahedron with face $F \in \mathcal{F}_h^\Gamma$ and apex p_0 ; $\hat{T} = \text{conv}(\hat{p}_0, \hat{p}_1, \hat{p}_2, \hat{p}_3)$ the reference tetrahedron $\hat{p}_0 = (0, 0, 1)$, $\hat{p}_1 = 0$, $\hat{p}_2 = e_1$, $\hat{p}_3 = e_2$, $\hat{F} = \{z = 0\} \cap \hat{T}$, inward unit normal $\hat{\nu} = e_3$; $x = A\hat{x} + a$ the affine map with $\hat{F} \mapsto F$, $\det A > 0$. Let $\nu = -n_F$ be the inward unit normal of T on F and h_T , $\hat{h} = 1$ the heights over F , $\hat{F}$. For $\hat{v}$ vanishing on $\hat{F}$, let $v := (\det A)^{-1} A \hat{v} \circ A^{-1}$ (Piola) and $\hat{g} := \partial_{\hat{\nu}} \hat{v}|_{\hat{F}}$.

Lemma 1.12. (a) *The Piola map preserves continuity, zero boundary values, piecewise- P_k structure on affinely corresponding meshes, and $\text{div } v = (\det A)^{-1} \widehat{\text{div}} \hat{v} \circ A^{-1}$. It maps the Alfeld split of $\hat{T}$ onto that of T .* (b) $\int_F \mu_i \mu_j \partial_\nu v \, dA = (\hat{h}/h_T)^2 A \int_{\hat{F}} \mu_i \mu_j \hat{g} \, d\hat{A}$ for all polynomial weights; in particular for $\mu_i \mu_j$ and for 1. (c) $\|\nabla v\|_T \leq (\det A)^{-1/2} \|A\| \|A^{-1}\| \|\hat{\nabla} \hat{v}\|_{\hat{T}}$.

Proof. (a) standard (A constant; barycentre $\mapsto$ barycentre). (b) On $\hat{F}$, $\hat{\nabla} \hat{v} = \hat{g} \otimes \hat{\nu}$, so $\partial_\nu v = (\det A)^{-1} A \hat{g} (A^{-\top} \hat{\nu} \cdot \nu)$ with $A^{-\top} \hat{\nu} = |A^{-\top} \hat{\nu}| \nu$ (it is normal to F and points into T). Nanson: $dA = \det A |A^{-\top} \hat{\nu}| \, d\hat{A}$. So the factor is $|A^{-\top} \hat{\nu}|^2$, and $|A^{-\top} \hat{\nu}| = |\nabla \lambda_0| \hat{h} = \hat{h}/h_T$ (λ_0 the barycentric of the apex). Barycentrics are affine invariants. (c) $\nabla v = (\det A)^{-1} A (\hat{\nabla} \hat{v}) A^{-1}$. $\square$

Since $\partial_n = -\partial_\nu$ on F , $\mathcal{M}_F(v) = -\frac{1}{2} \sum_{ij} \ell_{ij}^2 \int_F \mu_i \mu_j \partial_\nu v$; signs are absorbed by choosing $\pm v$.

Proposition 1.13 (Joint criterion). *Let $\hat{\mathcal{Y}}$ be a space of continuous piecewise polynomial, divergence-free fields on (a subdivision of) $\hat{T}$ vanishing on $\partial \hat{T}$, and $\hat{\mathcal{Y}}^1 := \{\hat{v} \in \hat{\mathcal{Y}} : \int_{\hat{F}} \hat{g} = 0\}$. Suppose the joint edge-moment map*

$$\hat{E} : \hat{\mathcal{Y}}^1 \rightarrow (\mathbb{R}^2)^3, \quad \hat{v} \mapsto \left(\int_{\hat{F}} \mu_1 \mu_2 \hat{g}, \int_{\hat{F}} \mu_1 \mu_3 \hat{g}, \int_{\hat{F}} \mu_2 \mu_3 \hat{g} \right)$$

is onto, with right inverse of norm $\|\hat{R}\|$ (from the Euclidean norm on $\mathbb{R}^6$ to $\|\hat{\nabla}\cdot\|_{L^2(\hat{T})}$). Then, for every tetrahedron T with face F on which the corresponding (Piola) space is admissible in Y_h^1 , and every set of real weights $w = (w_{12}, w_{13}, w_{23}) \neq 0$ and unit $t \parallel F$, there is v supported in T with $\|\nabla v\| = 1$ and

$$t \cdot \frac{1}{2} \sum_{i < j} w_{ij} \int_F \mu_i \mu_j \partial_\nu v \geq \frac{|w| (\hat{h}/h_T)^2 (\det A)^{1/2}}{2 \|A\| \|A^{-1}\|^2 \|\hat{R}\|}.$$

For the sphere ($w_{ij} = \ell_{ij}^2$, $|w| \geq (\text{diam } F)^2$) and a shape-regular T this is $\geq \kappa_0 (\text{diam } F)^{5/2}$, $\kappa_0 = c(\text{shape reg.})/\|\hat{R}\|$: (M^3) holds with $\omega_F = T$, $K_{\text{ov}} = 1$.

Proof. $s := A^{-1}t$ is parallel to $\hat{F}$. Take $\hat{v} := \hat{R}((w_{ij}s/|w|^2)_{i < j})$, so $\sum_{i < j} w_{ij} \hat{E}_{ij}(\hat{v}) = s$ and $\|\hat{\nabla} \hat{v}\| \leq \|\hat{R}\| \|s\|/|w| \leq \|\hat{R}\| \|A^{-1}\|/|w|$. By Lemma 1.12(b), $\frac{1}{2} \sum w_{ij} \int_F \mu_i \mu_j \partial_\nu v = \frac{1}{2} (\hat{h}/h_T)^2 A s = \frac{1}{2} (\hat{h}/h_T)^2 t$; by (c) and normalisation the claim follows. Shape regularity bounds $\|A\| \lesssim \text{diam } T$, $\|A^{-1}\| \lesssim (\text{diam } T)^{-1}$, $\det A \gtrsim (\text{diam } T)^3$, $h_T \lesssim \text{diam } T \lesssim \text{diam } F$; the powers combine to $(\text{diam } F)^{2-2+3/2-1+2} = (\text{diam } F)^{5/2}$. $\square$

No compactness over shapes is used: $\hat{\mathcal{Y}}^1$ and $\hat{E}$ are fixed, and the shape enters only through A and w . The weights may have any signs (used in Part B).

Theorem 1.14 ((M3³) on Alfeld splits, $k \geq 4$). *Let $\mathcal{T}_h$ be the Alfeld refinement of a shape-regular macro mesh of Ω_h and $k \geq 4$. Then (M^3) holds with $\omega_F = T_F$ (the macro-tetrahedron containing F), $K_{\text{ov}} = 1$ and $\kappa_0 = c(\text{shape reg.}) \hat{\kappa}$, $\hat{\kappa} := 1/\|\hat{R}\|$.*

Proof. Let $\hat{\mathcal{Y}}_k$ be the continuous piecewise- P_k divergence-free fields on the Alfeld split of $\hat{T}$ vanishing on $\partial \hat{T}$. Exact rational computation (`alfeld_witness.py` 4, `log_alfeld_k4.log`; Bernstein-Bézier basis on the 35 interior domain points, 105 unknowns, 80 divergence equations): $\text{rank div} = 79$ (onto the mean-zero P_3^{disc}), $\dim \hat{\mathcal{Y}}_4 = 26$, and the 8 functionals ($\int_{\hat{F}} \hat{g}$ and the three edge moments, 2 tangential components each) have rank 8 on $\hat{\mathcal{Y}}_4$. Hence $\hat{E}$ is onto on $\hat{\mathcal{Y}}_4^1$ ($\dim = 24$). For $k \geq 4$, $\hat{\mathcal{Y}}_4 \subset \hat{\mathcal{Y}}_k$. By Lemma 1.12(a) the Piola image of $\hat{\mathcal{Y}}_k^1$ under the map onto T_F , extended by zero, lies in Y_h^1 : it is continuous and piecewise P_k on the Alfeld refinement, vanishes on $\partial T_F \supset F$, is divergence free, has $\int_F \partial_n v = 0$ by (b), and T_F meets Γ_h only in F (Lemma 1.1), so the other constraints and moments vanish. Apply Prop. 1.13. $\square$

Value (NUMERICAL, floating point): with $\|\hat{\nabla}\cdot\|_{L^2(\hat{T})}$, the singular values of $\hat{E}$ on $\hat{\mathcal{Y}}_4^1$ are 0.240, 0.175, 0.123, 0.069, 0.0211, 0.0109, so $\hat{\kappa} = 0.0109$, $\|\hat{R}\| = 91.7$ (`k4.constant.py`).

Proposition 1.15 ((M3³) on any mesh, $k \geq 9$). *Let $K = K_F$, λ_0 its barycentric vanishing on F , $B := (\lambda_1 \lambda_2 \lambda_3)^2$. For a unit $\tau \parallel F$ and a quadratic q on F put $\psi := \lambda_0^2 B q (\nu \times \tau)$, q written as the same polynomial in $\lambda_1, \lambda_2, \lambda_3$ as in μ_1, μ_2, μ_3 , and $v := \text{curl } \psi \in P_9^3$, extended by zero. Then $v \in Y_h$, $\partial_\nu v|_F = -2|\nabla \lambda_0|^2 B q \tau$, and with $Q_w := \sum_{i < j} w_{ij} \mu_i \mu_j$, $q := Q_w - \int_F B Q_w / \int_F B$ one has $v \in Y_h^1$ and*

$$\frac{1}{2} \sum w_{ij} \int_F \mu_i \mu_j \partial_\nu v = -|\nabla \lambda_0|^2 \tau \int_F B (Q_w - \bar{Q}_w)^2, \quad \frac{1}{|F|} \int_F B (Q_w - \bar{Q}_w)^2 \geq \frac{|w|^2}{8316000}.$$

Hence (M^3) holds for $k \geq 9$ on every shape-regular mesh, with $\omega_F = K_F$, $K_{\text{ov}} = 1$, and any real weights.

Proof. ψ vanishes to second order on ∂K , so $v = \text{curl } \psi$ vanishes on ∂K , is continuous after extension by zero and divergence free. $\text{curl}(f\Phi) = \nabla f \times \Phi + f \text{curl } \Phi$ with $f = \lambda_0^2 Bq$; on F only $2\lambda_0 \partial_\nu \lambda_0 \dots$ survives after one normal derivative: $\partial_\nu v|_F = 2|\nabla \lambda_0|^2 Bq \nu \times (\nu \times \tau) = -2|\nabla \lambda_0|^2 Bq \tau$. Mean zero by the choice of the constant. The identity is then $\int B(Q_w - \bar{Q}_w)Q_w = \int B(Q_w - \bar{Q}_w)^2$. The lower bound: $\frac{1}{|F|} \int B(Q_w - \bar{Q}_w)^2 = w^\top \mathcal{V} w$ with $\mathcal{V}$ the B -weighted covariance of the three edge bubbles, computed exactly (`bubble_variance.py`): $\mathcal{V} = \frac{1}{1188000}I - \frac{1}{2772000}(J - I)$, eigenvalues $1/831600$ (twice) and $1/8316000$. By scaling (q has coefficients $O(|w|)$ in the barycentrics), $\|\nabla v\|_K \leq \|D^2 \psi\|_K \leq C(\text{shape})|w|(\text{diam } K)^{-1/2}$, while the moment has modulus $|\nabla \lambda_0|^2 |F| w^\top \mathcal{V} w \geq c(\text{shape})|w|^2/8316000$ ($|\nabla \lambda_0| = 1/h_K$). The ratio is $\geq c|w|(\text{diam } F)^{1/2}$, i.e. $\geq c(\text{diam } F)^{5/2}$ for $w_{ij} = \ell_{ij}^2$. $\square$

Proposition 1.16 ($k = 3$: one macro-element is not enough). *For $k = 3$ on the Alfeld split: $\dim \hat{\mathcal{Y}}_3 = 6$ (div onto mean-zero P_2^{disc}), $\dim \hat{\mathcal{Y}}_3^1 = 4$, and the image of $\hat{E}$ on $\hat{\mathcal{Y}}_3^1$ is the line spanned by $(\hat{E}_{12}, \hat{E}_{13}, \hat{E}_{23}) = (\hat{p}_1 - \hat{p}_2, \hat{p}_3 - \hat{p}_1, \hat{p}_2 - \hat{p}_3)$ (exact, `alfeld_k3.log`). Consequently, for a single macro-element field on T_F , the physical moment $\mathcal{M}_F(v)$ is a multiple of $\sum_{\text{cyc}} \ell_{ij}^2 e_{ij}$ (e_{ij} the cyclically oriented edge vectors), which is nonzero iff F is not equilateral; and one direction per face only. So (M^3) cannot be verified with $\omega_F = T_F$ for $k = 3$, not even for a single direction on equilateral faces.*

Proof. The rank statement is the exact computation. By Lemma 1.12(b), $\mathcal{M}_F = cA \sum \ell_{ij}^2 \hat{e}_{ij} = c \sum \ell_{ij}^2 e_{ij}$. With $a+b+c=0$ the cyclic edge vectors, $|a|^2 a + |b|^2 b + |c|^2 c = (|a|^2 - |c|^2)a + (|b|^2 - |c|^2)b$, zero iff $|a| = |b| = |c|$. In fact, symbolically (for a general triangle), $\sum_{\text{cyc}} \ell_{ij}^2 e_{ij} = \pm 12|F| R_{\pi/2}(x_c - o_F)$ with x_c the centroid and $R_{\pi/2}$ the in-plane rotation: the $k = 3$ macro-element sees exactly the centroid–circumcentre defect of td:faces(d) , rotated. $\square$

$k = 3$, larger patches (*NUMERICAL, exact rational ranks; `k3_patches.py`, `k3_equilateral_tests.py`, `k3_octa_weights.py`*). On vertex fans of m macro-tetrahedra $\text{conv}(q, z, a_j, a_{j+1})$ sharing an interior edge $[z, q]$, with the true circumsphere weights, the tangential total-moment map on $Y^1(\text{fan})$ has rank 2 for $m = 4, 5, 6$ (regular rings on a paraboloid), for an octahedral fan of *equilateral* faces with off-axis apex, and for a regular-tetrahedron fan of equilateral faces. It has rank 0 for the octahedral fan with apex on the axis, where the four interior macro-faces at $[z, q]$ lie in two planes (a “singular edge”, the analogue of (M1)); there the rank is restored by unequal weights ((1, 2, 3): rank 2). So (M^3) for $k = 3$ plausibly holds with fan patches under a non-singularity condition at interior edges, but this is not proved; Thm 1.10 is conditional for $k = 3$.

2 Part B: general smooth obstacles in 3D

2.1 Setting

Assumption 2.1 (Domain). $\Omega \subset \mathbb{R}^3$ bounded, connected, $\partial\Omega = \Gamma \sqcup \Sigma$; $\Gamma = \Gamma_1 \sqcup \dots \sqcup \Gamma_m$ closed connected embedded surfaces of class C^3 (class C^{k+3} where stated), each a whole component of $\partial\Omega$; $\Sigma \neq \emptyset$ a finite union of closed polyhedral surfaces (whole components), carrying strong data.

N is the unit normal of Γ into Ω , $n = -N$, d the signed distance (positive in Ω), $\mathfrak{h}_y(X, Y) = -\langle D_X N, Y \rangle$ the second fundamental form w.r.t. N (so $\mathfrak{h} > 0$ where Ω is locally convex; for the exterior of the unit ball $\mathfrak{h} = -I$), $\kappa_\infty := \sup |\mathfrak{h}|$.

Standard facts (fixed surface only; tubular neighbourhood theorem, regularity of d , implicit function theorem, compactness): (T) there is $r > 0$, $r\kappa_\infty \leq \frac{1}{2}$, with $\Psi(y, s) = y + sN(y)$ a C^2 diffeomorphism $\Gamma \times (-r, r) \rightarrow \mathcal{T}_r$, $d(\Psi(y, s)) = s$, $\pi = x - d\nabla d \in C^2$, $\nabla d = N \circ \pi$, $dx = J dA ds$

with $J = \det(I - sS_y) \in [\frac{1}{4}, \frac{9}{4}]$, $\mathcal{T}_r$ at positive distance from Σ , components $2r$ apart. (Gr) there is $a_0 \leq r/4$ with $\kappa_\infty a_0 \leq \frac{1}{40}$ such that in the frame $(y; \tau_1, \tau_2, N(y))$ with coordinates $(\xi, \eta) \in \mathbb{R}^2 \times \mathbb{R}$ and $Q'_y = (-2a_0, 2a_0)^3$: $\Gamma \cap Q'_y$ is the graph of $f_y \in C^3$, $\Omega \cap Q'_y = \{\eta > f_y\}$, $f_y(0) = 0$, $\nabla f_y(0) = 0$, $D^2 f_y(0) = \mathfrak{h}_y$, $|\nabla f_y(\xi)| \leq 2\kappa_\infty |\xi| \leq \frac{1}{5}$, $|f_y(\xi)| \leq \kappa_\infty |\xi|^2$, $\|D^3 f_y\| \leq C_\Gamma$.

Assumption 2.2 (Inscribed surfaces). (G1) $\Gamma_h = \bigcup_i \Gamma_{h,i}$, each $\Gamma_{h,i}$ a closed triangulated polyhedral surface whose vertices lie on Γ_i ; faces have diameter in $[c_0 h_\Gamma, h_\Gamma]$ and minimal angle $\geq \theta_0$. (G2) $\pi|_{\Gamma_h} : \Gamma_h \rightarrow \Gamma$ is a bijection. (G3) $h_\Gamma \leq h_*$, small.

(G2) is the standard lift condition of surface finite elements; it excludes folded triangulations and is automatic for convex inscribed polyhedra of the sphere (radial projection). It cannot be derived from (G1) alone.

2.2 Geometry

Lemma 2.3 (Faces on a general surface). *Let $F \in \mathcal{F}_h^\Gamma$ have vertices $z_i \in \Gamma$, barycentrics μ_i , $\ell := \text{diam } F \leq h_*$, edge vectors $e_{ij} = z_i - z_j$, centroid x_F , $y_F := \pi(x_F)$, $\mathfrak{h}_F := \mathfrak{h}_{y_F} \circ P_{y_F}$ (P the tangential projection). Then for $x \in F$, $x^* := \pi(x)$:*

- (a) $d(x) = \frac{1}{2} \sum_{i < j} \mathfrak{h}_F(e_{ij}, e_{ij}) \mu_i \mu_j + O(\ell^3)$; in particular $|d| \leq C\ell^2$;
- (b) $|n_F - n(x^*)| \leq C\ell$, where n_F is the unit normal of F with $n_F \cdot N(x^*) < 0$;
- (c) $\pi|_F$ is a C^2 diffeomorphism onto $\pi(F)$ with area Jacobian $1 + O(\ell^2)$;
- (d) $|w_F| \geq c(\theta_0) \ell^2 |\mathfrak{h}_{y_F}| - C\ell^3$, where $w_F := (\mathfrak{h}_F(e_{ij}, e_{ij}))_{i < j}$ and $|\mathfrak{h}|$ is the operator norm.

Proof. Frame (Gr) at $y := z_1$. $z_j = (\xi_j, f(\xi_j))$ with $|\xi_j| \leq \ell$; the vertical parts are $\leq \kappa_\infty \ell^2$, so the projected triangle $\tilde{F} = \text{conv}(\xi_j)$ has minimal angle $\geq \theta_0/2$ for h_* small, and F is the graph of $L := I_1 f$ over $\tilde{F}$ (same barycentrics). (a) $f = \frac{1}{2} \xi^\top H \xi + R_3$, $H = \mathfrak{h}_{z_1}$, $|R_3| \leq C\ell^3$ on $\tilde{F}$. For a quadratic form, $I_1 Q - Q = \sum_{i < j} \mu_i \mu_j Q(\xi_i - \xi_j)$ (variance identity, polarised). So $L - f = \frac{1}{2} \sum \mu_i \mu_j (\xi_i - \xi_j)^\top H (\xi_i - \xi_j) + O(\ell^3)$. For $x = (\xi, f(\xi) + \delta)$, $d(x) = \delta N_3 + O(\delta^2) = \delta + O(\ell^4)$ since $N_3 = (1 + |\nabla f|^2)^{-1/2} = 1 + O(\ell^2)$. Finally $\xi_i - \xi_j = P_{z_1} e_{ij}$, and replacing (z_1, P_{z_1}) by (y_F, P_{y_F}) changes $\mathfrak{h}(e, e)$ by $O(\ell)|e|^2$ ($\Gamma \in C^3$). (b) $n_F \parallel (\nabla L, -1)$, $N(x^*) \parallel (-\nabla f(\xi^*), 1)$; $|\nabla L - \nabla f| \leq C\ell/\sin(\theta_0/2)$ on $\tilde{F}$ (P1 interpolation), $|\xi^* - \xi| \leq C\ell^3$. (c) $D\pi(x) = (I - dS)^{-1} P_{x^*}$; restricted to the plane of F it has Jacobian $(n_F \cdot N(x^*)) (1 + O(d)) = 1 + O(\ell^2)$ by (b), (a). Injectivity on F : $\pi|_F$ is a small C^1 perturbation of the vertical projection, which is injective. (d) The map $H \mapsto (H(u_{ij}, u_{ij}))_{i < j}$, $u_{ij} = P e_{ij}/|P e_{ij}|$, from symmetric forms on $T_{y_F} \Gamma$ to $\mathbb{R}^3$ is invertible (three pairwise non-parallel directions) with inverse bounded by $C(\theta_0)$; $|P e_{ij}| \geq c\ell$. $\square$

For the unit sphere seen from $\Omega = R \setminus \bar{D}$: $\mathfrak{h} = -I$, $d(x) = -(1 - |x|)$ and (a) is Lemma 1.6 with $O(\ell^4)$.

Lemma 2.4 (Normal graph; definition of Ω_h). *Under (G1)–(G3), $\Gamma_h = \{y + \eta_h(y)N(y) : y \in \Gamma\}$ with $\eta_h := d \circ (\pi|_{\Gamma_h})^{-1}$ Lipschitz, piecewise C^2 , $\|\eta_h\|_\infty \leq Ch_\Gamma^2$, $\|\nabla_\Gamma \eta_h\|_\infty \leq Ch_\Gamma$. Define*

$$\Omega_h := (\Omega \setminus \mathcal{T}_{r/2}) \cup \{x \in \mathcal{T}_{r/2} : d(x) > \eta_h(\pi(x))\}.$$

Then Ω_h is a bounded polyhedral domain, $\partial\Omega_h = \Sigma \sqcup \Gamma_h$, Ω_h lies on the N -side of each face (its outward normal on F is n_F of Lemma 2.3(b)), $\Omega_h \setminus \Omega = \{\eta_h \circ \pi < d \leq 0\}$, $\Omega \setminus \Omega_h = \{0 < d \leq \eta_h \circ \pi\}$, $|\Omega \Delta \Omega_h| \leq Ch_\Gamma^2$. Both differences are nonempty in general (e.g. near saddle points, where $\mathfrak{h}$ is indefinite, a single face crosses Γ).

Proof. Every point of a face is within h_Γ of a vertex, so $\Gamma_h \subset \mathcal{T}_{r/2}$. By (G2) η_h is well defined; its bounds follow from Lemma 2.3(a) and, on $\pi(F)$, $\nabla_\Gamma \eta_h = D(\pi|_F)^{-\top} \nabla_F d$ with $|\nabla_F d| = |N(x^*) - (N \cdot n_F)n_F| \leq Ch_\Gamma$ (b) and (c). Since $\pi(x + tN(\pi x)) = \pi(x)$, near F the set $\{d > \eta_h \circ \pi\}$ is the half-space side into which N points. The rest is as in `gd:lem:graph`. $\square$

Lemma 2.5 (Uniform Korn, trace, Friedrichs, Poincaré; connectedness). *The statements of `gd:lem:korn` hold on $\Omega_h \subset \mathbb{R}^3$ with constants independent of h .*

Proof. $\Phi_h(\Psi(y, s)) := \Psi(y, s + \chi(s)\eta_h(y))$ (χ as in `gd:lem:korn`) is a homeomorphism $\overline{\Omega} \rightarrow \overline{\Omega}_h$, identity near Σ , with $\|D\Phi_h - I\|_\infty + \|D\Phi_h^{-1} - I\|_\infty \leq Ch_\Gamma$: in (y, s) coordinates $D\hat{\Phi}_h = I + \begin{pmatrix} 0 & 0 \\ \chi \nabla_\Gamma \eta_h & \chi' \eta_h \end{pmatrix}$, and $D\Psi(y, s')D\Psi(y, s)^{-1} = I + O(|s - s'|)$. Then steps 4–6 of Lemma `lem:korn` verbatim (Korn's first inequality with zero trace on Σ , trace, Friedrichs and Poincaré on the fixed Lipschitz domain Ω , transferred). $\square$

2.3 Extension across Γ without vector potentials

Lemma 2.6 (Piola–Hestenes extension). *Let $\Gamma \in C^{k+3}$, $u \in C^{k+1}(\overline{\Omega})$ divergence free, $p \in C^k(\overline{\Omega})$. There are $\tilde{u} \in C^{k+1}(\mathcal{O})$, $\tilde{p} \in C^k(\mathcal{O})$, $\mathcal{O} := \Omega \cup \mathcal{T}_{r/2}$, with $\tilde{u} = u$, $\tilde{p} = p$ on Ω and $\operatorname{div} \tilde{u} = 0$ on $\mathcal{O}$. Moreover $\|\tilde{u}\|_{L^\infty(\Gamma_h)} \leq Ch_\Gamma^2$ and, with $F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$ (supported in $\overline{\Omega}_h \setminus \Omega$), $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$ for $v \in V_h^R$.*

Proof. Let $\omega := \iota_u \operatorname{vol}$, a closed C^{k+1} 2-form on $\overline{\Omega}$. On a collar $\Gamma_i \times [0, r)$, $\beta := \Psi^* \omega$ is closed and C^{k+1} up to $s = 0$ ($\Psi \in C^{k+2}$ since $N \in C^{k+2}$). In local coordinates (σ_1, σ_2) on Γ_i , $\beta = a d\sigma_1 \wedge d\sigma_2 + b_1 d\sigma_2 \wedge ds + b_2 ds \wedge d\sigma_1$. For $\lambda > 0$ let $R_\lambda(y, s) = (y, -\lambda s)$; then $R_\lambda^* \beta = a(\sigma, -\lambda s) d\sigma_1 \wedge d\sigma_2 - \lambda b_\alpha(\sigma, -\lambda s)(\dots)$. Choose distinct $\lambda_j \in (0, 1]$, $j = 1, \dots, k+3$, and c_j with $\sum_j c_j (-\lambda_j)^m = 1$ for $m = 0, \dots, k+2$ (Vandermonde), and set $\tilde{\beta} := \sum_j c_j R_{\lambda_j}^* \beta$ for $s \in (-r/2, 0)$. Then $\partial_s^m \partial_\sigma^\alpha$ of the coefficients of $\tilde{\beta}$ and β agree at $s = 0$ for $m + |\alpha| \leq k+1$ (for a this uses $m \leq k+1$, for b the shifted conditions $m+1 \leq k+2$), so the glued form is C^{k+1} on $\Gamma_i \times (-r/2, r)$; it is closed on each side (pull-backs commute with d) hence everywhere. Push forward by Ψ^{-1} and let $\tilde{u}$ be the vector field of the resulting closed 2-form: $\tilde{u} \in C^{k+1}$, $\operatorname{div} \tilde{u} = 0$. $\tilde{p}$: scalar Hestenes reflection. The bounds are those of `gd:lem:ext`: $\tilde{u}|_\Gamma = 0$ and $|d| \leq Ch_\Gamma^2$ on Γ_h ; $|(F, v)|$ by a one-dimensional Poincaré inequality along normal segments across $\{\eta_h \circ \pi < d \leq 0\}$ (width $\leq Ch_\Gamma^2$), the Jacobian J , Lemma 2.3(c) and Lemma 2.5. $\square$

This replaces the vector potential of `td:setting` (whose existence on a collar needs the zero-flux condition and elliptic regularity) by an elementary construction; it also works in 2D. For the geometric lemmas only $\tilde{u} \in C^2$ near Γ is needed.

2.4 Uniform Bogovskiĭ constant

Lemma 2.7 (Uniform star-shaped cover in 3D). *There are $N_B, r_*, d_*, m_* > 0$ depending only on Ω such that for $h_\Gamma \leq h_*$, Ω_h admits a cover as in `gd:lem:bog` (which is dimension-free, with Galdi's star-shaped constant $c_0(d/r)^3(1+d/r)$). Hence the Bogovskiĭ (continuous inf-sup) constant of Ω_h is bounded independently of h .*

Proof. Let $a := a_0$ of (Gr), frames at $y_1, \dots, y_{N_1} \in \Gamma$ such that the sets $\Gamma \cap \{|\xi|_\infty < a/4\}$ (in the frame at y_j) cover Γ ; $Q_j := (-a, a)^3$, $U_j^h := \Omega_h \cap Q_j$.

1. $\Gamma_h \cap Q_j$ is a Lipschitz graph. With $y(\xi) = (\xi, f(\xi))$, the points of Γ_h over $\Gamma \cap Q_j'$ are $y(\xi) + \eta_h(y(\xi))N(y(\xi))$, with horizontal coordinate $\Xi(\xi) = \xi - \eta_h \nabla f / \sqrt{1 + |\nabla f|^2}$. By Lemma 2.4,

$\|D\Xi - I\| \leq Ch_\Gamma$ and $|\Xi(\xi) - \xi| \leq Ch_\Gamma^2$, so Ξ is a bi-Lipschitz homeomorphism of $(-2a, 2a)^2$ onto a set containing $(-\frac{3}{2}a, \frac{3}{2}a)^2$; over it Γ_h is the graph of $f_h := (f + \eta_h/\sqrt{1+|\nabla f|^2}) \circ \Xi^{-1}$ with $\text{Lip}(f_h) \leq \frac{1}{5} + Ch_\Gamma \leq L := \frac{1}{4}$ and $|f_h| \leq 2\kappa_\infty a^2 + Ch_\Gamma^2 \leq b_0 := \frac{a}{10}$ on $(-a, a)^2$. No other part of Γ_h enters $(-\frac{3}{2}a, \frac{3}{2}a)^2$: points of Γ_h are within Ch_Γ^2 of their projection on Γ , and $\Gamma \cap Q'_j$ is the graph part.

2. $U_j^h = \{(\xi, \eta) \in Q_j : \eta > f_h(\xi)\}$, by the connectedness argument of gd:lem:cover, step 2 (the point $(0, 0, a/2)$ is in Ω_h , $(0, 0, -a/2)$ is not).

3. *Star-shaped w.r.t. every point of $B_j := B((0, 0, \frac{3}{4}a), \frac{a}{8})$.* For $y \in B_j$, $x \in U_j^h$, $z_t = (1-t)y + tx$: $|\xi_x - \xi_y| \leq (\sqrt{2} + \frac{1}{8})a < 1.54a$ and

$$\eta_{z_t} - f_h(\xi_{z_t}) \geq t(\eta_x - f_h(\xi_x)) + (1-t)(\frac{5}{8}a - b_0 - 1.54La) \geq t(\eta_x - f_h(\xi_x)) + (1-t)0.14a > 0;$$

the box constraints are convex; $\overline{B_j} \subset U_j^h$; $\text{diam } U_j^h \leq 2\sqrt{3}a$.

4. *Interior pieces, covering, overlaps.* $\Omega_* := \{x \in \Omega : \text{dist}(x, \Gamma) > a/4\}$ is an h -independent connected Lipschitz domain (homeomorphic to Ω via the collar map), $\Omega_* \subset \Omega_h$, and a finite union of domains star-shaped w.r.t. balls (Galdi Lemma II.1.3; locator unverified as in the paper). If $x \in \Omega_h$ and $\text{dist}(x, \Gamma) < a/2$, then $x = \pi(x) + dN$ with $\pi(x)$ in the $a/4$ -patch of some y_j , and in that frame $|\xi_x|_\infty \leq \frac{a}{4} + \frac{a}{2} \cdot \frac{1}{5} < a$, $|\eta_x| \leq \frac{a}{10} + \frac{a}{2} < a$, so $x \in U_j^h$. Ordering chart pieces first: $U_j^h \cap \Omega_* \supset y_j + \{|\xi|_\infty < \frac{a}{2}, \frac{a}{2} < \eta < a\}$ (these points are at distance $\geq (\frac{a}{2} - b_0)/\sqrt{1+L^2} > \frac{a}{4}$ from Γ), of volume $a^3/2$. The rest is gd:lem:cover. $\square$

Consequently the domain part of (IS3) is uniform on general smooth obstacles; the discrete part has exactly the status of td:open item 3.

2.5 Test field, fluxes and the global mean

Lemma 2.8 (Divergence-free field with normal trace q). *Let $\Gamma \in C^{k+3}$, $q \in C^{k+1}(\Gamma)$, $\int_\Gamma q = 0$, $\bar{q}_i$ its mean on Γ_i . Let $\phi_i \in C^{k+2, \alpha}(\Gamma_i)$ solve $\Delta_\Gamma \phi_i = q - \bar{q}_i$ (Schauder theory on the closed C^{k+3} surface), χ a cutoff ($\chi = 1$ near 0, $\chi = 0$ on $[r/4, r)$), and on the product $\Gamma \times (-r, r)$ (volume $dA ds$) let $V := (-\chi'(s)\nabla_\Gamma \phi_i, \chi(s)q)$. Let w_0 be the Piola push-forward of V by Ψ (defined by $\iota_{w_0} \text{vol} = (\Psi^{-1})^* \iota_V (dA \wedge ds)$). Then $\text{div } w_0 = J^{-1} \chi'(s) \bar{q}_i$, $w_0 = qN$ on Γ , $w_0 \in C^{k+1}$. Correcting the $\bar{q}_i$ -part with Bogovskii fields on balls in $\{d > r/16\}$ exactly as in gd:lem:w (dimension-free) gives $w \in C^{k+1}$ on a neighbourhood of $\bar{\Omega}$, $\text{div } w = 0$, $w = 0$ near Σ , $w|_\Gamma = qN = -qn$, with the bounds (c) of gd:lem:w.*

Proof. $d(\iota_{w_0} \text{vol}) = (\Psi^{-1})^*((\text{div}_{\text{prod}} V) dA \wedge ds)$ and $\Psi^* \text{vol} = J dA \wedge ds$; $\text{div}_{\text{prod}} V = -\chi' \Delta_\Gamma \phi_i + \chi' q = \chi' \bar{q}_i$. At $s = 0$, $D\Psi = \text{id}_{T\Gamma} \oplus (\partial_s \mapsto N)$, $J = 1$, $\chi' = 0$. Regularity: $\nabla_\Gamma \phi_i \in C^{k+1, \alpha}$, $D\Psi, J \in C^{k+1}$. For $m = 1$ and piecewise constant q , $\phi_i = 0$. $\square$

The *discrete* leak field is then the interpolate-and-correct field of td:vphi, $v_\varphi := I_h w - ab - v_0 \in Z_h$; its proof uses only $\int_{\Gamma_h} w \cdot n = \int_{\Omega_h} \text{div } w - \int_\Sigma w \cdot \nu = 0$, boundary tetrahedra of size $\leq Ch_\Gamma$, the face bubbles and (IS3). No global stream function or vector potential is needed, so the multiply-connected complications of gd:lem:curlint disappear in 3D.

Lemma 2.9 (Fluxes). *For $v \in Z_h$, $\sum_i \Phi_i(v) = 0$, $\Phi_i(v) := \int_{\Gamma_{h,i}} v \cdot n$. If $m \geq 2$ and $\Gamma \in C^{k+3}$, $v \mapsto (\Phi_i(v))_i$ maps Z_h onto $\{\sum_i x_i = 0\}$ for small h .*

Proof. As gd:lem:flux, with v_α from Lemma 2.8 ($q = \alpha_i$ on Γ_i) and td:vphi: $\Phi_i(v_\alpha) = -|\Gamma_i| \alpha_i + O(h_\Gamma^2 + h^k)|\alpha|$ by Lemma 2.3(b),(c). $\square$

2.6 Main theorem on general obstacles

Theorem 2.10. *Assume Assumptions 2.1, 2.2, $(M0^3)$ (with ∂R replaced by Σ), $(IS3)$ with β independent of h , (D) , and $u \in C^{k+1}(\bar{\Omega})$, $p \in C^k(\bar{\Omega})$, $\Gamma \in C^{k+3}$. In CNS^* use the global mean $\bar{p}_\Gamma(q) = |\Gamma_h|^{-1} \int_{\Gamma_h} q$ and $\bar{p} := |\Gamma|^{-1} \int_\Gamma p$, $G := \|p - \bar{p}\|_{L^2(\Gamma)}$. Then Theorem `td:main` (i), (ii) and Corollary `td:ns` hold verbatim on Ω_h ; the results using v_φ additionally need $p|_\Gamma \in C^{k+1}$. Proposition `gd:prop:Gglobal` (the constant is global; the per-component G is not an admissible error indicator when $m \geq 2$) and Remark `gd:rem:percomp` (per-component correction inconsistent) hold verbatim. $\Omega \subset \Omega_h$ is not used anywhere.*

Proof. Go through the proof of `td:main` and replace each ingredient:

- `td:faces` $\rightarrow$ Lemma 2.3: sagitta $O(h_\Gamma^2)$, $|n_F - n(x^*)| \leq Ch_\Gamma$, Jacobian $1 + O(h_\Gamma^2)$ (only these are used, as noted after `td:faces`);
- `td:tools(a)` $\rightarrow$ Lemma 2.5; `td:tools(b)` and the extension $\rightarrow$ Lemma 2.6; `td:tools(c),(d)` are mesh-only (face bubble on any $F \in \mathcal{F}_h^\Gamma$, Ω_h connected);
- `td:bog` $\rightarrow$ Lemma 2.7;
- `td:transposed(ii)`: $(\nabla \tilde{u})^\top n_F = n(x^*)(\partial_n u(x^*) \cdot (n_F - n(x^*))) + O(h_\Gamma^2) = O(h_\Gamma)$ (Lemma `lem:wall` holds on any C^1 surface), so `td:G` holds;
- `td:w` $\rightarrow$ Lemma 2.8 with $q = p - \bar{p}$; `td:vphi` verbatim;
- `td:cancel`: Γ_h is a closed triangulated surface, so every edge lies in exactly two faces; $|n_F - n_{F'}| \leq Ch_\Gamma$ for adjacent faces (both within Ch_Γ of n at nearby points); $|w - (w \cdot n_F)n_F| \leq Ch_\Gamma$ on F since $w = qN + O(d)$ near Γ . Steps 1–5 unchanged;
- Cor. `cor:GSerr` and the consistency step of Thm `thm:D` use only $\sum_i \Phi_i(v) = 0$ on Z_h (Lemma 2.9); the global constant is forced as in `gd:cor:GSerr`, and Prop. `gd:prop:Gglobal` follows from the lower bound of Thm A with $\tilde{u} = 0$;
- step 4 of Thm `thm:C`: $(\tilde{p} - \bar{p})n_F + v_\varphi = q(x^*)(n_F - n(x^*)) + O(h_\Gamma^2 + h^k) = O(h_\Gamma)$, energy-dual bound (`td:nocancel2d`); step 2 of Thm `thm:B(i)`: $\tilde{u} = d \partial_N u(x^*) + O(d^2)$ is tangential to Γ up to $O(h_\Gamma^4)$, $v_\varphi = qN + O(h_\Gamma^2 + h^k)$;
- the Riemann sums $\int_{\Gamma_h}(\cdot) \approx \int_\Gamma(\cdot)$ (e.g. $\mathcal{R}(v_\varphi) = G^2 + O(h_\Gamma^2 + h^k)$) use (G2) and Lemma 2.3(c).

All integrations by parts are on the polyhedron Ω_h , where $\tilde{u} \in C^2(\bar{\Omega}_h)$, $\tilde{p} \in C^1(\bar{\Omega}_h)$; F lives on $\Omega_h \setminus \Omega$; $\Omega \setminus \Omega_h$ is never used. $\square$

Theorem 2.11 (Curvature-weighted lower bound on general obstacles). *Under the assumptions of Theorem 2.10, on an Alfeld-refined mesh with $k \geq 4$ or on any mesh with $k \geq 9$, let $W := \partial_N u|_\Gamma$ and $|\mathfrak{h}|$ the pointwise operator norm of the second fundamental form. Then for CNS^* ($\gamma \geq \gamma_2$) and GS ($\gamma \geq \gamma_0$), if $\mathfrak{h}W \not\equiv 0$ and $h_\Gamma \leq h_1$,*

$$\|\nabla(\tilde{u} - u_h)\|_{\Omega_h} \geq c h_\Gamma^{3/2} \| |\mathfrak{h}| W \|_{L^2(\Gamma)} - Ch_\Gamma^2,$$

with c depending on shape regularity, θ_0 , and $\hat{\kappa}$ (Alfeld) resp. $\lambda_{\min}(\mathcal{V}) = 1/8316000$ (bubble).

Proof. Lemmas 1.1 (with the remark in its proof), 1.2–1.4 and Cor. 1.5 are unchanged. In Lemma 1.7 replace Lemma 1.6 by Lemma 2.3(a): $\tilde{u} = dW(x^*) + O(d^2)$ with $d = \frac{1}{2} \sum w_{ij}^F \mu_i \mu_j + O(\ell^3)$, so $\mathcal{M}_F(v) := \frac{1}{2} \sum_{i < j} w_{ij}^F \int_F \mu_i \mu_j \partial_n v$ and the error is still $Ch_\Gamma^3 \|\partial_n v\|_{L^1(F)}$, summing to $Ch_\Gamma^{5/2} \|\nabla v\|$. Prop. 1.13 and Prop. 1.15 allow arbitrary real weights and give a witness with $t \cdot \mathcal{M}_F(v_{F,t}) \geq c|w^F|(\text{diam } F)^{1/2} \geq ch_\Gamma^{5/2} |\mathfrak{h}_{y_F}| - Ch_\Gamma^{7/2}$ (Lemma 2.3(d)). In Lemma 1.9 take $v_F := |\mathfrak{h}_{y_F}| |W_F^\parallel| v_{F,t_F}$; the Riemann sum is now for $\int_\Gamma |\mathfrak{h}|^2 |W|^2$ ($|\mathfrak{h}|$ is Lipschitz), and $\|\nabla v\|^2 \leq \sum_F |\mathfrak{h}_{y_F}|^2 |W_F|^2$. The rest is identical. $\square$

On flat parts of Γ the faces lie on Γ and the chordal bump vanishes; the theorem says the $h_\Gamma^{3/2}$ loss is driven by $|\mathfrak{h}| |\partial_N u|$, as expected. Whether $\mathfrak{h}$ could be replaced by a smaller quantity (e.g. only the part of $\mathfrak{h}$ seen by the mesh) is not examined.

3 What is not proved

1. (M3³) for $k = 3$ (Alfeld): single macro-elements provably fail (Prop. 1.16); fan patches work in exact tests except at a singular-edge configuration. No proof.
2. (M3³) for $4 \leq k \leq 8$ on non-split meshes (where (IS3) is anyway unknown), and for Worsey–Farin splits (boundary faces subdivided; the witness of Prop. 1.15 still works for $k \geq 9$ with the weight $-\frac{1}{2}|x - o_F|^2$ restricted to the sub-face, since only the nonzero Hessian of the weight is used).
3. The upper bounds (Thm td:main, Thm 2.10) keep the td:open item 3 caveat (uniformity of the discrete split-mesh inf-sup constant); the domain part is now uniform for general obstacles (Lemma 2.7).
4. (G2) is an assumption, as in surface FEM.
5. Galdi locators (Lemma II.1.3, III.3.1, Thm III.3.3) are used as in the paper and inherit its VERIFY flags.
6. No 3D Stokes computation was run; the only numerics are the exact/rational rank computations and one floating-point singular value computation.

Files (notes/v6/td_sharp/)

- `alfeld_witness.py k` (exact; logs `alfeld_k3.log`, `alfeld_k4.log`): reference Alfeld macro-element, $\dim \hat{\mathcal{Y}}_k$, rank of divergence, rank of the 8 face functionals. Run: `python3 alfeld_witness.py 4` (4 s).
- `k4_constant.py` (log `k4_constant.log`): singular values of $\hat{E}$ in the H^1 -seminorm (floating point).
- `bubble_variance.py` (log): exact covariance matrix $\mathcal{V}$ of Prop. 1.15.
- `alfeld_patch.py`: general exact patch code (independent implementation; single-tet cross-check of the $k = 3$ and $k = 4$ ranks on a non-reference shape).
- `k3_patches.py`, `k3_equilateral_tests.py`, `k3_octa.py`, `k3_octa_weights.py`, `k3_trace.py`, `k3_constraints.py`, `k3_face_moment.py` and logs: the $k = 3$ exploration.